external transform space

$$V_{24} = \left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{!5}\right)\left(\frac{4\pi}{35}\right)^1\left(\frac{4\pi}{18}\right)^2\left(\frac{4\pi}{32}\right)^3\left(\frac{4\pi}{8}\right)^4$$

$$\rho_{\text{Leech}} = \left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{!5}\right)\left(\frac{4\pi}{35}\right)^1\left(\frac{4\pi}{18}\right)^2\left(\frac{4\pi}{32}\right)^3\left(\frac{4\pi}{8}\right)^4$$

$$5\left(\frac{4\pi}{2}\right)\left(\cos\left(\frac{7}{5}\right)\right)^2 = \frac{8}{e^{\phi_4}}$$ mass

$$2\left(\frac{5}{\sqrt{7}}\right)^2\left(\cos\left(\frac{5}{2}i\right)\right)^2\left(\cos\left(\frac{7}{5}\right)\right)^2 = \frac{32}{e^{\phi_3}}$$ temperature

$$\left(\frac{5}{\sqrt{7}}\right)\left(\frac{3^{-1/3}}{W_{We}}\right) = \frac{18}{e^{\phi_2}}$$ charge

$$\left(\sinh\left(\sinh\left(\frac{1}{7}\right)\right)\right)^{-1} = \frac{35}{e^{\phi_1}}$$ length

$$\left(\frac{4\pi}{4}\right)\left(\sinh\left(\frac{1}{4}\right)\right)^2 = \frac{!5}{e^{\phi_0}}$$ time

$\phi_4 = 2.1764268381757\ldots$ $\quad 2(3-1)^2 = 8$

$\phi_3 = 1.4167869859079\ldots$ $\quad 2(3+1)^2 = 32$

$\phi_2 = 1.8755459671396\ldots$ $\quad 2(3\pm 0)^2 = 18$

$\phi_1 = 1.6162591817564\ldots$ $\quad 7\times 5 = 35 \quad \Gamma(5) = 24$

$\phi_0 = 5.3912583683231\ldots$ $\quad !5 = 44 \quad 5! = 120$

$m_p = 2.1764268381757\ldots\times 10^{-8}$ kg $\quad m_p = \phi_4\times 10^{-8}$ kg

$T_p = 1.4167869859079\ldots\times 10^{32}$ K $\quad T_p = \phi_3\times 10^{32}$ K

$q_p = 1.8755459671396\ldots\times 10^{-18}$ C $\quad q_p = \phi_2\times 10^{-18}$ C

$l_p = 1.6162591817564\ldots\times 10^{-35}$ m $\quad l_p = \phi_1\times 10^{-35}$ m

$t_p = 5.3912583683231\ldots\times 10^{-44}$ s $\quad t_p = \phi_0\times 10^{-44}$ s

$$A_{\text{ext}}B_{\text{ext}}\left(1 + A_{\text{int}}\boxtimes\right) \qquad \boxtimes = \frac{l_p m_p}{{q_p}^2}\times\frac{\text{coulomb}^2}{\text{meter kilogram}}$$

$$\frac{1}{x} + x + \frac{x^3}{2\pi} = \left(i^i\right)^{-\frac{4\pi}{8}} - \frac{m_p}{\text{kilogram}}$$

$ж_1 = 0.0854245431533304\ldots$

$ж_2 = 3.66756753485501\ldots$

$ж_3 = -1.87649603900417\ldots + 4.06615262615972\ldots\, i$

$ж_4 = -1.87649603900417\ldots - 4.06615262615972\ldots\, i$

$ж_r = 4.47826244916751\ldots$ $\quad ж_3 = ж_r\, e^{ж_\theta i}$

$ж_\theta = 2.00316562310924\ldots$ $\quad ж_4 = ж_r\, e^{-ж_\theta i}$

$ж_1ж_2ж_3ж_4 = 2\pi$ $\quad -Im(ж_3) = Im(ж_4)$

$ж_1 + ж_2 + ж_3 + ж_4 = 0$ $\quad Im(ж_1) = 0$

$-ж_1 - ж_2 - ж_3 - ж_4 = 0$ $\quad Im(ж_2) = 0$

${ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2 = -4\pi$

$-{ж_1}^2 - {ж_2}^2 - {ж_3}^2 - {ж_4}^2 = 4\pi$ $\quad \frac{ж_1 + ж_2}{ж_3 + ж_4} = -1$

${ж_1}^3 + {ж_2}^3 + {ж_3}^3 + {ж_4}^3 = 6\pi a$

${ж_1}^4 + {ж_2}^4 + {ж_3}^4 + {ж_4}^4 = 8\pi(\pi - 1)$ $\quad \frac{{ж_1}^2{ж_2}^2}{2\pi} = \frac{2\pi}{{ж_3}^2{ж_4}^2}$

$\frac{1}{ж_1} + \frac{1}{ж_2} + \frac{1}{ж_3} + \frac{1}{ж_4} = a$

$ж_1ж_2 + ж_1ж_3 + ж_1ж_4 + ж_2ж_3 + ж_2ж_4 + ж_3ж_4 = 2\pi$

$ж_1ж_2ж_3 + ж_1ж_2ж_4 + ж_2ж_3ж_4 = 2\pi a$

$ж_1 + ж_2 = -2\,Re(ж_3) = -2\,Re(ж_4)$

$ж_1 - ж_2 = Re(ж_1) - Re(ж_2)$

$ж_3 + ж_4 = 2\,Re(ж_3) = 2\,Re(ж_4)$

$ж_3 - ж_4 = 2i\,Im(ж_3) = -2i\,Im(ж_4)$

$Re(ж_1) + Re(ж_2) + Re(ж_3) + Re(ж_4) = 0$

$Im(ж_1) + Im(ж_2) + Im(ж_3) + Im(ж_4) = 0$

$Re(ж_3)^2 + Im(ж_3)^2 = {ж_r}^2$

$Re(ж_3)^2 + Im(ж_4)^2 = {ж_r}^2$ $\quad ж_3ж_4 = {ж_r}^2$

$Re(ж_4)^2 + Im(ж_3)^2 = {ж_r}^2$ $\quad ж_1ж_2 = \frac{2\pi}{{ж_r}^2}$

$Re(ж_4)^2 + Im(ж_4)^2 = {ж_r}^2$

$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2 = {ж_r}^4$

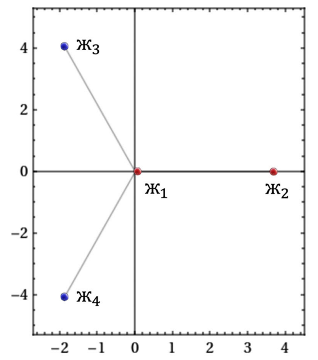
$(a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2 = {ж_r}^4$

$a = Re(ж_3) \quad b = Im(ж_3) \quad c = Re(ж_4) \quad d = Im(ж_4)$

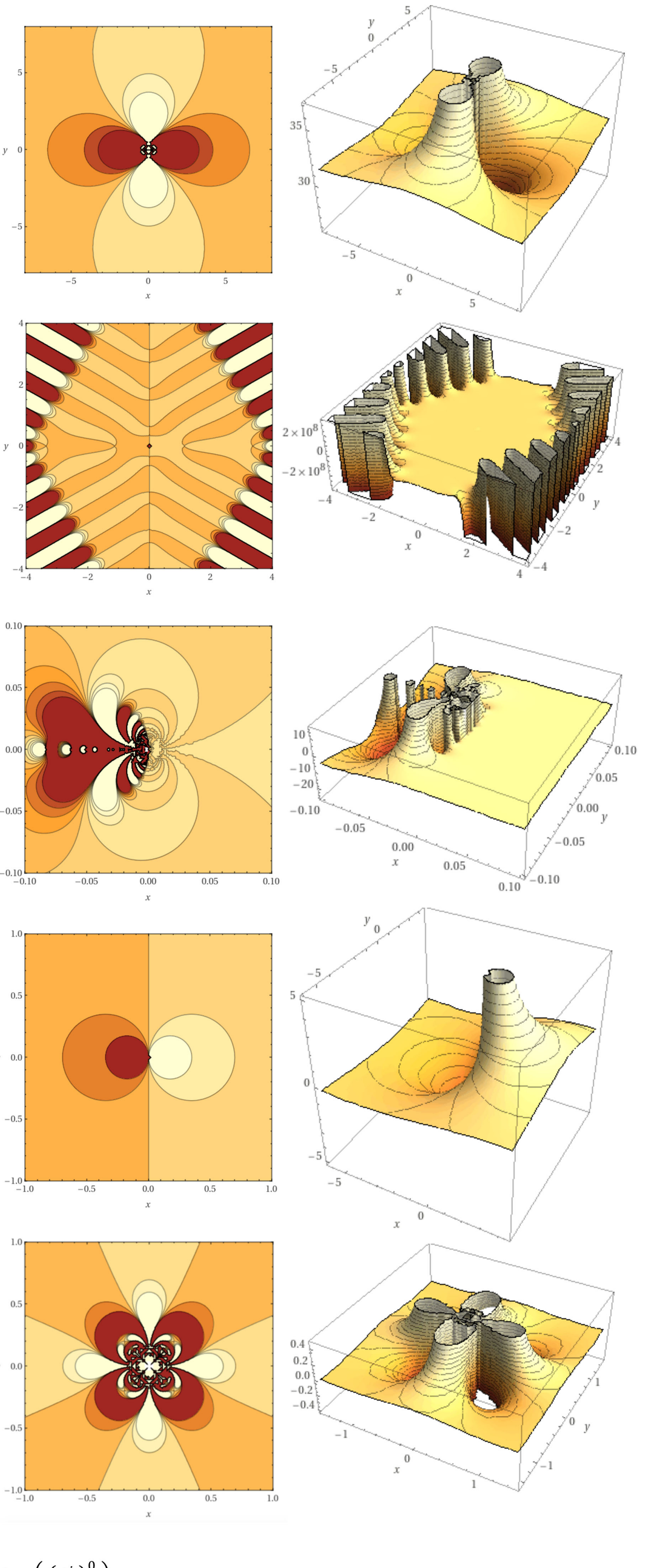
$a = Re(ж_4) \quad b = Im(ж_4) \quad c = Re(ж_3) \quad d = Im(ж_3)$

$a = Im(ж_3) \quad b = Re(ж_3) \quad c = Im(ж_4) \quad d = Re(ж_4)$

$a = Im(ж_4) \quad b = Re(ж_4) \quad c = Im(ж_3) \quad d = Re(ж_3)$



$$\sum_{n=1}^{4} ж_n = 0 \qquad \prod_{n=1}^{4} ж_n = 2\pi \qquad \sum_{n=1}^{4} {ж_n}^2 = -4\pi$$



$\log\left(\left(i^i\right)^0\right) = ж_1 + ж_2 + ж_3 + ж_4$

$\log\left(\left(i^i\right)^4\right) = -ж_1ж_2ж_3ж_4$

$\log\left(\left(i^i\right)^8\right) = {ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2$

$\log\left(\left(i^i\right)^4\right) = -1\,ж_1ж_2ж_3ж_4$

$\log\left(\left(i^i\right)^8\right) = -2\,ж_1ж_2ж_3ж_4$

$\log\left(\left(i^i\right)^{12}\right) = -3\,ж_1ж_2ж_3ж_4$

$\log\left(\left(i^i\right)^{16}\right) = -4\,ж_1ж_2ж_3ж_4$

$\log\left(\left(i^i\right)^8\right) = 1\left({ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2\right)$

$\log\left(\left(i^i\right)^{16}\right) = 2\left({ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2\right)$

$\log\left(\left(i^i\right)^{24}\right) = 3\left({ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2\right)$

$\log\left(\left(i^i\right)^{32}\right) = 4\left({ж_1}^2 + {ж_2}^2 + {j_3}^2 + {ж_4}^2\right)$

$S_{n-1}(r) = \frac{2\pi^{n/2}}{\Gamma\left(\frac{n}{2}\right)} r^{n-1}$

$V_n(r) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)} r^n$

$\frac{S_{n+1}(r)}{V_n(r)} = 2\pi r$

$\frac{S_{n-1}(r)}{V_n(r)} = \frac{n}{r}$

$\left(i^i\right)^{-1} = \frac{s^4\,{W_{We}}^4}{8}$

$\left(i^i\right)^{-2} = e^{\pi}$

$\left(i^i\right)^{-4} = e^{2\pi}$

$\left(i^i\right)^{-6} = e^{3\pi}$

$\left(i^i\right)^{-8} = e^{4\pi}$

$$\sum_{n=0}^{\infty}\frac{1}{n!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \frac{1}{6!} + \frac{1}{7!} + \frac{1}{8!} + \cdots = e$$

$$\sum_{n=0}^{\infty}\frac{(-1)^n}{n!} = \frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} + \frac{1}{8!} - \cdots = \frac{1}{e}$$

$$\sum_{n=0}^{\infty} x^n = x^0 + x^1 + x^2 + x^3 + x^4 + x^5 + x^6 + x^7 + \cdots = \frac{1}{1-x}$$

$$\sum_{n=0}^{\infty} (-x)^n = x^0 - x^1 + x^2 - x^3 + x^4 - x^5 + x^6 - x^7 + \cdots = \frac{1}{1+x}$$

$\left[Li_2\left(\frac{1}{\varphi_i}\right) + Li_2(\varphi_i)\right] = \frac{(4\pi)^2}{288}$

$i\left[Li_2\left(\frac{1}{\varphi_i}\right) - Li_2(\varphi_i)\right] = V_{\text{fe}}$

$i\left[Li_2\left(\frac{1}{\varphi_i^2}\right) - Li_2(\varphi_i^2)\right] = \frac{2V_{\text{fe}}}{3}$

$i\left[Li_2\left(\frac{1}{\varphi_i^3}\right) - Li_2(\varphi_i^3)\right] = 0$

$i\left[Li_2\left(\frac{1}{\varphi_i^4}\right) - Li_2(\varphi_i^4)\right] = -\frac{2V_{\text{fe}}}{3}$

$i\left[Li_2\left(\frac{1}{\varphi_i^5}\right) - Li_2(\varphi_i^5)\right] = -V_{\text{fe}}$

$i\left[Li_2\left(\frac{1}{\varphi_i^6}\right) - Li_2(\varphi_i^6)\right] = 0$

$i\left[Li_2\left(\frac{1}{i}\right) - Li_2(i)\right] = 2K$

$i\left[Li_2\left(\frac{1}{i^2}\right) - Li_2(i^2)\right] = 0$

$i\left[Li_2\left(\frac{1}{i^3}\right) - Li_2(i^3)\right] = -2K$

$i\left[Li_2\left(\frac{1}{i^4}\right) - Li_2(i^4)\right] = 0$

$\left[Li_2\left(\frac{1}{i}\right) + Li_2(i)\right] = -\frac{1}{6}\left(\frac{4\pi}{8}\right)^2$

$Li_2(1) - Li_2(-1) = \left(\frac{4\pi}{8}\right)^2$

$Li_2(-1) = -\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{8}\right)$

$Li_2(1) = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{4}\right)$

$\sum_{n=1}^{\infty}\frac{1}{n^2} = Li_2(1)$

$\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} = Li_2(-1)$

$\sum_{n=1}^{\infty}\frac{1}{(2n+1)^2} = \left(\frac{4\pi}{16}\right)\left(\frac{4\pi}{8}\right)$

$\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2} = K$

$\sum_{n=1}^{\infty}\frac{\sin\left(n\frac{\pi}{2}\right)}{n^2} = K$

$\sum_{n=1}^{\infty}\frac{\sin\left(n\frac{\pi}{3}\right)}{n^2} = G_{\text{Gi}}$

$\sum_{n=1}^{\infty}\frac{\cos\left(n\frac{\pi}{3}\right)}{n^2} = \left(\frac{4\pi}{\Gamma(5)}\right)^2$

$\sum_{n=1}^{\infty}\frac{\cos\left(n\frac{\pi}{2}\right)}{n^2} = -\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{32}\right)$

$\sum_{n=0}^{\infty}\binom{n}{k} = 2^n$

$\sum_{n=1}^{\infty}\frac{1}{n^s} = \zeta(s) \qquad \sum_{n=0}^{\infty}\frac{H_n}{n^2} = 2\zeta(3)$

$\sum_{n=0}^{\infty}\varphi^{-n} = \varphi^2 \qquad \sum_{n=0}^{\infty}\psi^{-n} = \psi^3$

$\sum_{n=1}^{\infty}\frac{F_n}{F_{2n}} = \mathcal{L}_{RL}$

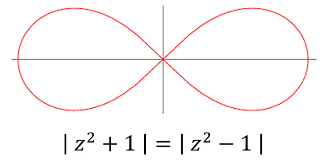
$\sum_{n=1}^{\infty}\frac{1}{F_n} = \mathcal{F} = \sqrt{5}\sum_{n=1}^{\infty}\frac{1}{\varphi^n - (-\varphi)^n}$

$\sum_{n=1}^{\infty}\frac{1}{L_n} = \mathcal{L}_{RL} = \sum_{n=1}^{\infty}\frac{1}{(-\varphi)^{-n} + \varphi^n}$

$\sum_{n=0}^{\infty}\frac{x^n}{n!} = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \cdots = e^x$

lemniscate

$(x^2 + y^2)^2 = (x^2 - y^2)$



$|z^2 + 1| = |z^2 - 1|$

defines points equidistant from roots 0, 1, and −1

$Li_2(\varphi_i) = \left(\frac{4\pi}{\Gamma(5)}\right)^2 + G_{\text{Gi}}\, i$

$Li_2\left(\frac{1}{\varphi_i}\right) = \left(\frac{4\pi}{\Gamma(5)}\right)^2 - G_{\text{Gi}}\, i$

$Li_2(i) = -\left(\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{32}\right)\right) + K\, i$

$Li_2\left(\frac{1}{i}\right) = -\left(\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{32}\right)\right) - K\, i$

$\log\left(\left(i^i\right)^{-\pi/2}\right) = \left(\frac{4\pi}{8}\right)^2$

$\log\left(\left(i^i\right)^{-32\pi}\right) = (4\pi)^2$

$\log\left(\left(i^i\right)^8\right) = -4\pi$

$\log\left(\left(i^i\right)^4\right) = -2\pi$

$\log\left(\left(i^i\right)^2\right) = -\left(\frac{4\pi}{4}\right)$

$\log\left(\left(i^i\right)^{1/1}\right) = -\left(\frac{4\pi}{8}\right)$

$\log\left(\left(i^i\right)^{1/2}\right) = -\left(\frac{4\pi}{16}\right)$

$\log\left(\left(i^i\right)^{1/3}\right) = -\left(\frac{4\pi}{\Gamma(5)}\right)$

$\log\left(\left(i^i\right)^{1/4}\right) = -\left(\frac{4\pi}{32}\right)$

$\frac{S}{2} = \sin\left(\frac{3\pi}{14}\right) + 1$

$\frac{S}{2} = \cos\left(\frac{4\pi}{14}\right) + 1$

$e^{\pi i} = -1$

$e^{\pi} = \left(i^{-i}\right)^2$

$i^{-i} = \frac{s^4\,{W_{\text{We}}}^4}{8}$

$e^{\pi} = \frac{s^8\,{W_{\text{We}}}^8}{8^2}$

$i^{-i} = 2L^4\,{W_{\text{We}}}^4$

$e^{\pi} = 4L^8\,{W_{\text{We}}}^8$

$\left(i^i\right)^{-4\pi/8} = \frac{s^{2\pi}\,{W_{\text{We}}}^{2\pi}}{\sqrt{8^{\pi}}}$

$\left(i^i\right)^{-2\pi} = \frac{s^{8\pi}\,{W_{\text{We}}}^{8\pi}}{8^{2\pi}}$

$W_{We} = \frac{\sqrt{\sqrt{8}}}{s}\, e^{\left(\frac{4\pi}{32}\right)}$

$W_{\text{We}} = \frac{1}{L}\left(\frac{i^{-i}}{2}\right)^{1/4}$

$W_{\text{We}} = \frac{1}{L}\left(\frac{e^{\pi}}{4}\right)^{1/8}$

$W_{\text{We}} = \frac{1}{L}\left(\frac{e^{2\pi}}{16}\right)^{1/16}$

$W_{\text{We}} = \frac{1}{L}\left(\frac{e^{4\pi}}{8\cdot 32}\right)^{1/32}$

$e^{\zeta'(0)} = \frac{1}{\sqrt{2\pi}}$

$e^{\beta'(0)} = \frac{s}{\sqrt{4\pi}}$

$\omega_1 = \frac{\Gamma^3\left(\frac{1}{3}\right)}{4\pi}\varphi_i$

$\omega_2 = \frac{\Gamma^3\left(\frac{1}{3}\right)}{4\pi}$

$s = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{2\pi}}$

$L = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{8\pi}}$

$L_1 = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{32\pi}}$

$L_2 = \frac{2\pi}{\sqrt{2}}\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma^2\left(\frac{1}{4}\right)}$

$C_U = 2\pi\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma^2\left(\frac{1}{4}\right)}$

$G_{Ga} = \frac{\Gamma^2\left(\frac{1}{4}\right)}{2\pi\sqrt{2\pi}}$

$\frac{r_n}{r_e} = \left(\frac{4\pi}{\Gamma(5)}\right)^2$

$\frac{r_n}{r_+} = K$

$G_{\text{Gi}} = V_{\text{ideal tetrahedron}}$

$2G_{\text{Gi}} = V_{\text{fe}}$

$2K = 2\pi\, C_d$

$4K = 2\pi\log(D_d)$

$4K = V_{\text{Whitehead link}}$

$4K = V_{\text{ideal octahedron}}$

$8K = V_{\text{Borromean rings}}$

$\sin\left(\frac{3\pi}{14}\right) = \cos\left(\frac{4\pi}{14}\right)$

$\sin\left(\frac{4\pi}{14}\right) = \cos\left(\frac{3\pi}{14}\right)$

$\left(\frac{4\pi}{8}\right) = L\,L_2$

$\left(\frac{4\pi}{16}\right) = L_1L_2 \qquad \cos\left(\frac{4\pi}{8}\right) = 0$

$\left(\frac{4\pi}{32}\right) = \frac{L\,L_1L_2}{s} \qquad \sin\left(\frac{4\pi}{8}\right) = 1$

$\int_0^{\pi} Cl_2{}^2(x)\,dx = \zeta(4)$

$\int_0^{2\pi/12} -4\log\left|e^{-xi} - e^{xi}\right| = V_{\text{fe}}$

$\int_0^{2\pi/6} -6\log\left|e^{-xi} - e^{xi}\right| = V_{\text{fe}}$

$\int_0^{1/2}\Gamma(1+x)\,\Gamma(1-x)\,dx = \frac{4K}{2\pi}$

$\int_0^{1}\Gamma(1+x)\,\Gamma(1-x)\,dx = \frac{8K}{2\pi}$

$\int_0^1\int_0^1\frac{1}{1+x^2y^2} = K$

$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\frac{1}{2}2\pi}$

$\int_{-\infty}^{\infty} e^{-x^4}\,dx = \sqrt{\frac{L}{2}\sqrt{2\pi}}$

$\int_{-1}^{1}\sqrt{1+x^2}\,dx = P_{up}$

$\int_{-1}^{1}\sqrt{1-x^2}\,dx = \left(\frac{4\pi}{8}\right)$

$\int_{-\infty}^{\infty}\frac{1}{1+x^2}\,dx = \pi$

$\int_{-\infty}^{\infty}\frac{1}{1+x^4}\,dx = \frac{4\pi}{\sqrt{32}}$

$\int_0^1\frac{1}{\sqrt{1+x^4}}\,dx = \frac{s}{\sqrt{32}}$

$\int_0^1\frac{1}{\sqrt{1-x^4}}\,dx = \frac{s}{4}$

$\int_0^1\sqrt{1-x^4}\,dx = \frac{s}{6}$

$\int_0^1\frac{x^2}{\sqrt{1-x^4}}\,dx = \frac{\pi}{s}$

$\int_0^{1/2}\int_0^{x} 6\sqrt{x^2+y^2}\,dy\,dx = \frac{P_{up}}{8}$

$\int_0^{\infty} 2^{-x}\,dx = \frac{1}{\log(2)}$

$\int_0^1\frac{\log(x)}{e^x}\,dx = \gamma$

$\int_0^{\infty}\log\left(\log\left(\frac{1}{x}\right)\right)dx = \gamma$

Combinatorics

$V_{\text{unit sphere}} = \left(\frac{4\pi}{3}\right)$

$V_{\text{Gabriel's horn}} = \left(\frac{4\pi}{4}\right)$

$V_{\text{pseudosphere}} = \left(\frac{4\pi}{6}\right)$

$V_{\text{exp trumpet}} = \left(\frac{4\pi}{8}\right)$

$V_{\text{cos spindle}} = \left(\frac{4\pi}{4}\right)\left(\frac{4\pi}{8}\right)$

$A_{\text{one pedal}} = 1$

$A_{\text{unit lemniscate}} = 2$

$A_{\text{exp trumpet}} = \pi\,P_{up}$

$A_{\text{cos spindle}} = 2\pi\,P_{up}$

$A_{\text{unit circle}} = 2\pi$

$A_{\text{unit sphere}} = 4\pi$

$A_{\text{pseudosphere}} = 4\pi$

$A_{\text{unit cardioid}} = 6\pi$

$L_{\text{unit wave}} = \frac{4\pi}{s} + s$

$L_{\text{unit cardioid}} = 8$

$L_{\text{unit cardioid cusp normal}} = \sqrt{\frac{32}{18}}$

$\alpha_F = \lim_{n\to\infty}\frac{d_n}{d_{n+1}}$

$\delta_F = \lim_{n\to\infty}\frac{a_{n-1} - a_{n-2}}{a_n - a_{n-1}}$

3D sphere packing densities

$\rho_{\text{cubic lat}} = \left(\frac{4\pi}{\Gamma(5)}\right)$

$\rho_{\text{hex lat}} = \frac{2}{\sqrt{3}}\left(\frac{4\pi}{\Gamma(5)}\right)$

$\rho_{\text{tet lat}} = \frac{\sqrt{3}}{2}\left(\frac{4\pi}{32}\right)$

$\rho_{\text{BCC}} = \sqrt{3}\left(\frac{4\pi}{32}\right)$

$\rho_{\min} = \frac{1}{\sqrt{2}}\left(\frac{4\pi}{18}\right)$

$\rho_{\text{FCC}} = \frac{4\pi}{\sqrt{288}}$

$\rho_{\text{HCP}} = \frac{4\pi}{\sqrt{288}}$

densest hypersphere lattice packings

$\delta_2 = \sqrt{3}\left(\frac{4\pi}{\Gamma(5)}\right)$

$\delta_3 = \sqrt{2}\left(\frac{4\pi}{\Gamma(5)}\right)$

$\delta_4 = \left(\frac{4\pi}{16}\right)^2$

$\delta_5 = \sqrt{2}\left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{4}\right)$

$\delta_6 = \sqrt{3}\left(\frac{4\pi}{16}\right)\left(\frac{4\pi}{18}\right)\left(\frac{4\pi}{32}\right)$

$\delta_7 = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{35}\right)\left(\frac{4\pi}{8}\right)$

$\delta_8 = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{16}\right)^3$

max density sphere packings

$\rho_{2D} = \sqrt{\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{8}\right)}$

$\rho_{3D} = \frac{4\pi}{\sqrt{288}}$

$\rho_{4D} = \left(\frac{4\pi}{16}\right)^2$

$\rho_{8D} = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{16}\right)^3$

factors of 288

1	288
2	144
3	96
4	72
6	48
8	**36**
9	**32**
12	24
16	**18**

$\Gamma(\varphi + 1) = \Gamma(\varphi - 1)$

$\Gamma(\varphi_i + 1) = -\Gamma(\varphi_i - 1)$

$\Gamma\left(-\frac{1}{\varphi} + 1\right) = \Gamma\left(-\frac{1}{\varphi} - 1\right)$

$\Gamma\left(\frac{1}{\varphi_i} + 1\right) = -\Gamma\left(\frac{1}{\varphi_i} - 1\right)$

fixed point equation for $i^{i^{i^{\cdot^{\cdot}}}}$

$x = i^x$

$x = \cos\left(\frac{4\pi}{8}x\right) + i\sin\left(\frac{4\pi}{8}x\right)$

$\log(x) = \left(\frac{4\pi}{8}\right)xi$

$x = e^{\left(\frac{4\pi}{8}\right)xi}$

fixed point equation for $\pm C_{\text{CFP}}$

$e^x(x - 1) = e^{-x}(x + 1)$

$x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

$\coth(x) = x$

$\text{csch}(C_{\text{CFP}}) = L_{\text{LL}}$

$\text{csch}(-C_{\text{CFP}}) = -L_{\text{LL}}$

$\sinh(C_{\text{CFP}}) = \frac{1}{L_{\text{LL}}}$

$\sinh(-C_{\text{CFP}}) = -\frac{1}{L_{\text{LL}}}$

$\zeta\left(\tan\left(\frac{4\pi}{32}\right) + 1\right) = \zeta\left(\cot\left(\frac{4\pi}{32}\right) - 1\right)$

$\zeta\left(-\tan\left(\frac{4\pi}{32}\right) - 1\right) = \zeta\left(-\cot\left(\frac{4\pi}{32}\right) + 1\right)$

fixed point equation for D_{Do}

$\cos(x) = x$

$x = Re\left(e^{xi}\right)$

fixed point equation for Ω

$x = e^{-x}$

$x = -\log(x)$

$x = \log\left(\frac{1}{x}\right)$

$\prod_{n=1}^{\infty}\left(1 - \left(-\frac{1}{\varphi^2}\right)^n\right) = F_{\text{FF}}$ $\quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

$\prod_{p\text{ prime}}\left(\frac{p^2+1}{p^2-1}\right) = \frac{5}{2}$ $\quad \Gamma\left(-\frac{1}{2}\right) = -\sqrt{4\pi}$

$\prod_{p\text{ prime}}\left(\frac{p^4+1}{p^4-1}\right) = \frac{7}{6}$ $\quad \Gamma\left(\frac{1}{2}\right)\Gamma\left(-\frac{1}{2}\right) = -2\pi$

$\prod_{p\text{ prime}}\left(\frac{p^s+1}{p^s-1}\right) = \frac{\zeta(s)^2}{\zeta(2s)}$ $\quad \Gamma\left(\frac{3}{2}\right) = \sqrt{\frac{4\pi}{16}}$

$\prod_{p\text{ prime}}\left(1 + \frac{1}{p^s}\right) = \frac{\zeta(s)}{\zeta(2s)}$ $\quad B\left(\frac{1}{2},\frac{3}{2}\right) = \left(\frac{4\pi}{8}\right)$

$\prod_{p\text{ prime}}\left(1 - \frac{1}{p^s}\right) = \frac{1}{\zeta(s)}$ $\quad B\left(\frac{1}{2},\frac{1}{2}\right) = \pi$

$B\left(\frac{1}{2},\frac{1}{4}\right) = s$

$\prod_{p\text{ prime}}\left(1 + \frac{1}{p(p-1)}\right) = \left(\frac{35}{4\pi}\right)\left(\frac{18}{4\pi}\right)\left(\frac{8}{4\pi}\right)^2\zeta(3)$

$(1+i)!\,(1-i)! = \frac{4\pi}{e^{\pi} - e^{-\pi}}$

$(0+i)!\,(0-i)! = \frac{2\pi}{e^{\pi} - e^{-\pi}}$

$|\Gamma(0+i)|^2 = \frac{2\pi}{e^{\pi} - e^{-\pi}}$

$\left|\Gamma\left(\frac{1}{2}+i\right)\right|^2 = \frac{2\pi}{e^{\pi} + e^{-\pi}}$

$|\Gamma(1+i)|^2 = \frac{2\pi}{e^{\pi} - e^{-\pi}}$

$\lim_{n\to\infty}\frac{1}{\log(p_n)}\prod_{k=1}^{n}\frac{1}{1 - \frac{1}{p_k}} = e^{\gamma}$

$\lim_{n\to\infty}\log(p_n)\prod_{k=1}^{n}\frac{1}{1 + \frac{1}{p_k}} = \frac{6}{e^{\gamma}}\left(\frac{4\pi}{\Gamma(5)}\right)^2$

$\lim_{x\to 0}\frac{1}{x}\left(\frac{1}{\Psi(1-x)} - \frac{1}{\Psi(1+x)}\right) = \frac{6}{\gamma^2}\frac{(4\pi)^2}{288}$

$\lim_{n\to\infty}\frac{\pi^0}{\Gamma(1)} + \frac{\pi^1}{\Gamma(2)} + \frac{\pi^2}{\Gamma(3)} + \cdots + \frac{\pi^n}{\Gamma(n+1)} = e^{\pi}$

$\lim_{x\to 0}\frac{1}{x}\left(\frac{1}{\Gamma(1+x)} - \frac{1}{\Gamma(1-x)}\right) = 2\gamma$

$\lim_{x\to 0}\left(\frac{\zeta(1+x) + \zeta(1-x)}{2}\right) = \gamma$

$\lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \int_1^n\frac{1}{x}\,dx\right) = \gamma$

$\Gamma'(1) = -\gamma$

$s = 2L \qquad s = 4K(i)$

$L = 2L_1 \qquad C_d = \frac{2K}{2\pi}$

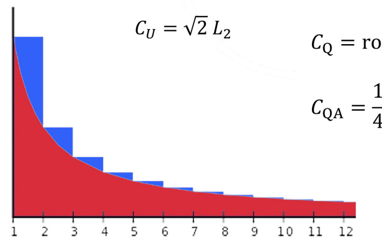
$L_1 = \frac{s}{4} \qquad D_d = e^{\left(\frac{4K}{2\pi}\right)}$

$L_2 = \frac{\pi}{s} \qquad D_d = e^{2C_d}$

$G_{Ga} = \frac{s}{2\pi} \qquad S = \log\left(\left(e^{\gamma}C_U\right)^2\right)$

$C_U = \sqrt{2}\,L_2 \qquad C_Q = \text{root of }\zeta\left(\frac{1}{2},\frac{x}{2}\right)$

$C_{QA} = \frac{1}{4}\zeta\left(\frac{3}{2},\frac{C_Q}{2}\right)$



$\varphi = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}}}}$

$\frac{1}{\varphi} = 0 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}}}}$

$C_{\text{CF}} = 0 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{1}{3 + \cfrac{1}{4 + \cfrac{1}{5 + \ddots}}}}}$

$C_{\text{R1}} = 0 + \cfrac{e^{-0\pi}}{1 + \cfrac{e^{-2\pi}}{1 + \cfrac{e^{-4\pi}}{1 + \cfrac{e^{-6\pi}}{1 + \cfrac{e^{-8\pi}}{1 + \ddots}}}}}$

$\frac{1}{1-e} = 0 + \cfrac{1}{1 + \cfrac{2}{2 + \cfrac{3}{3 + \cfrac{4}{4 + \cfrac{5}{5 + \ddots}}}}}$

$e = 2 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{2}{3 + \cfrac{3}{4 + \cfrac{4}{5 + \ddots}}}}}$

$\left(\frac{4\pi}{8}\right) = 1 + \cfrac{1}{\frac{1}{1} + \cfrac{1}{\frac{1}{2} + \cfrac{1}{\frac{1}{3} + \cfrac{1}{\frac{1}{4} + \cfrac{1}{\frac{1}{5} + \ddots}}}}}$

$\left(\frac{4\pi}{16}\right) = 0 + \cfrac{1}{1 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{7 + \cfrac{1}{9 + \ddots}}}}}$

$\left(\frac{16}{4\pi}\right) = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cfrac{7^2}{2 + \cfrac{9^2}{2 + \ddots}}}}}$

$\left(\frac{s^2}{4\pi}\right) = 2 + \cfrac{1^2}{4 + \cfrac{3^2}{4 + \cfrac{5^2}{4 + \cfrac{7^2}{4 + \cfrac{9^2}{4 + \ddots}}}}}$

$\left(\frac{4\pi}{4}\right) = 3 + \cfrac{1^2}{6 + \cfrac{3^2}{6 + \cfrac{5^2}{6 + \cfrac{7^2}{6 + \cfrac{9^2}{6 + \ddots}}}}}$

$18\left(\frac{2\pi}{s^2}\right) = 4 + \cfrac{1^2}{8 + \cfrac{3^2}{8 + \cfrac{5^2}{8 + \cfrac{7^2}{8 + \cfrac{9^2}{8 + \ddots}}}}}$

$K = 0 + \cfrac{1^4}{1 + \cfrac{1^4}{8 + \cfrac{3^4}{16 + \cfrac{5^4}{24 + \cfrac{7^4}{32 + \ddots}}}}}$

$2K = 1 + \cfrac{1\cdot 1}{3 + \cfrac{1\cdot 2}{1 + \cfrac{2\cdot 2}{3 + \cfrac{2\cdot 3}{1 + \cfrac{3\cdot 3}{3 + \ddots}}}}}$

$\frac{\pi}{2} = 1 + \cfrac{1\cdot 1}{1 + \cfrac{1\cdot 2}{1 + \cfrac{2\cdot 3}{1 + \cfrac{3\cdot 4}{1 + \cfrac{4\cdot 5}{1 + \ddots}}}}}$

$\frac{L}{2} = 1 + \cfrac{1\cdot 1}{2 + \cfrac{2\cdot 3}{2 + \cfrac{4\cdot 5}{2 + \cfrac{6\cdot 7}{2 + \cfrac{8\cdot 9}{2 + \ddots}}}}}$

$\cosh(x) = \frac{1}{2}\left(e^x + e^{-x}\right)$

$\sinh(x) = \frac{1}{2}\left(e^x - e^{-x}\right)$

$\cos(x) = \frac{1}{2}\left(e^{xi} + e^{-xi}\right)$

$\sin(x) = \frac{i}{2}\left(e^{xi} - e^{-xi}\right)$

$\cosh(x) = \frac{x^0}{0!} + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \cdots$

$\sinh(x) = \frac{x^1}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \frac{x^9}{9!} + \cdots$

$\cos(x) = \frac{x^0}{0!} - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots$

$\sin(x) = \frac{x^1}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots$

$\tan^{-1}(x) = \frac{x^1}{1} - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots$

$\log(1+x) = \frac{x^1}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots$

$\log(1-x) = -\frac{x^1}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \cdots$

$\log(x) = \frac{x^{-1}}{1} - \frac{x^{-2}}{2} + \frac{x^{-3}}{3} - \frac{x^{-4}}{4} + \frac{x^{-5}}{5} - \cdots$

$Li_s(x) = \frac{x^1}{1^s} + \frac{x^2}{2^s} + \frac{x^3}{3^s} + \frac{x^4}{4^s} + \frac{x^5}{5^s} + \cdots$

$\zeta(1) = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots = \infty$

$\zeta(2) = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{6}$

$\zeta(3) = \frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \cdots$

$\zeta(3) = \frac{1}{2}\int_0^{\infty}\frac{\log(x)\log(1-x)}{x(1-x)}\,dx$

$\zeta(3) = \frac{1}{2}\int_0^{\infty}\frac{x^2}{e^x - 1}\,dx \qquad \sqrt{\frac{\zeta(3)}{\zeta'(-2)}} = 2\pi i$

$\zeta(3) = \frac{2}{3}\int_0^{\infty}\frac{x^2}{e^x + 1}\,dx \qquad \frac{\zeta\left(\frac{3}{2}\right)}{\zeta\left(-\frac{1}{2}\right)} = -4\pi$

Pascal's triangle

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1

$\sqrt{288} = 16 + \cfrac{1}{1 + \cfrac{1}{32 + \cfrac{1}{1 + \cfrac{1}{32 + \cfrac{1}{1 + \ddots}}}}}$

natural numbers $= \binom{n+0}{1} = \frac{n}{1!}$

triangular numbers $= \binom{n+1}{2} = \frac{n(n+1)}{2!}$

tetrahedral numbers $= \binom{n+2}{3} = \frac{n(n+1)(n+2)}{3!}$

pentatope numbers $= \binom{n+3}{4} = \frac{n(n+1)(n+2)(n+3)}{4!}$

5 – simplex numbers $= \binom{n+4}{5} = \frac{n(n+1)(n+2)(n+3)(n+4)}{5!}$

$1^2 + 2^2 + 3^2 + 4^2 + 5^2 + \cdots + n^2 = \frac{n(n-1)(2n+1)}{3!}$

$1^3 + 2^3 + 3^3 + 4^3 + 5^3 + \cdots + n^3 = (1 + 2 + 3 + 4 + 5 + \cdots + n)^2$

$(1+x)^n = \frac{n^{\underline{0}}x^0}{0!} + \frac{n^{\underline{1}}x^1}{1!} + \frac{n^{\underline{2}}x^2}{2!} + \frac{n^{\underline{3}}x^3}{3!} + \frac{n^{\underline{4}}x^4}{4!} + \cdots + \frac{n^{\underline{n}}x^n}{n!}$

$2^n = \frac{n^{\underline{0}}}{0!} + \frac{n^{\underline{1}}}{1!} + \frac{n^{\underline{2}}}{2!} + \frac{n^{\underline{3}}}{3!} + \frac{n^{\underline{4}}}{4!} + \frac{n^{\underline{5}}}{5!} + \frac{n^{\underline{6}}}{6!} + \frac{n^{\underline{7}}}{7!} + \frac{n^{\underline{8}}}{8!} + \frac{n^{\underline{9}}}{9!} + \frac{n^{\underline{10}}}{10!} + \cdots$

$2^n = \frac{1}{0!} + \frac{n}{1!} + \frac{n(n-1)}{2!} + \frac{n(n-1)(n-2)}{3!} + \cdots$

$\frac{\pi}{4} = \frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \frac{1}{13} - \frac{1}{15} + \frac{1}{17} - \frac{1}{19} + \frac{1}{21} - \frac{1}{23} + \frac{1}{25}\cdots$

$\infty = \frac{2}{1}\,\frac{4}{3}\,\frac{6}{5}\,\frac{8}{7}\,\frac{10}{9}\,\frac{12}{11}\,\frac{14}{13}\,\frac{16}{15}\,\frac{18}{17}\,\frac{20}{19}\,\frac{22}{21}\,\frac{24}{23}\,\frac{26}{25}\,\frac{28}{27}\,\frac{30}{29}\,\frac{32}{31}\,\frac{34}{33}\,\frac{36}{35}\cdots$

$0 = \frac{2}{3}\,\frac{4}{5}\,\frac{6}{7}\,\frac{8}{9}\,\frac{10}{11}\,\frac{12}{13}\,\frac{14}{15}\,\frac{16}{17}\,\frac{18}{19}\,\frac{20}{21}\,\frac{22}{23}\,\frac{24}{25}\,\frac{26}{27}\,\frac{28}{29}\,\frac{30}{31}\,\frac{32}{33}\,\frac{34}{35}\,\frac{36}{37}\cdots$

$\frac{\pi}{2} = \prod_{n=1}^{\infty}\left(1 + \frac{1}{n}\right)^{(-1)^{n+1}} = \prod_{n=1}^{\infty}\left(\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right) = \left(\frac{2}{1}\cdot\frac{2}{3}\right)\left(\frac{4}{3}\cdot\frac{4}{5}\right)\cdots$

$\frac{L}{2} = \prod_{n=1}^{\infty}\left(1 + \frac{1}{2n}\right)^{(-1)^{n+1}} = \prod_{n=1}^{\infty}\left(\frac{4n-1}{4n-2}\cdot\frac{4n}{4n+1}\right) = \left(\frac{3}{2}\cdot\frac{4}{5}\right)\left(\frac{7}{6}\cdot\frac{8}{9}\right)\cdots$

$n! = \int_0^{\infty}(x-1)^n\,e^{-x}\,dx$

$!n = n!\sum_{k=0}^{n}\frac{(-1)^k}{k!}$

$!0 = 1$

$!1 = 0$

$\Gamma(s) = \int_0^{\infty}\frac{x^{s-1}}{e^x}\,dx$

$Q(1,x) = e^{-x}$

$P(1,x) = 1 - e^{-x}$

$n! = \Gamma(n+1)$

$\Gamma(n) = (n-1)!$

$\frac{1}{\Gamma(-n)} = 0$

$\zeta(-2n) = 0$

$\zeta(s) = Li_s(1)$

$\zeta(0) = -\frac{1}{2}\log(e)$

$\zeta'(0) = -\frac{1}{2}\log(2\pi)$

$\zeta(s) = \frac{1}{\Gamma(s)}\int_0^{\infty}\frac{x^{s-1}}{e^x - 1}\,dx$

$\zeta(s) = \prod_{p\text{ prime}}\frac{1}{1 - p^{-s}}$

$\log(ab) = \log(a) + \log(b)$

$\log(n) = \int_1^n\frac{1}{x}\,dx$

$\frac{d}{dx}\log(x) = \frac{1}{x}$

$\log(x) = -\log\left(\frac{1}{x}\right)$

$Li_1(x) = -\log(1-x)$

$Li_0(x) = \frac{x}{1-x}$

$Li_{-1}(x) = \frac{x}{(1-x)^2}$

$Li_{-2}(x) = \frac{x(1+x)}{(1-x)^3}$

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$

$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} = \prod_{k=1}^{n}\frac{n+k}{k}$

$(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^kb^{n-k}$

$|i| = 1 \qquad P + 1 = P^3$

$|\varphi_i| = 1 \qquad P - 1 = P^{-4}$

$i^4 = 1 \qquad \varphi = \frac{a}{b} = \frac{a+b}{a}$

$\varphi_i^6 = 1 \qquad \varphi_i = \frac{a}{b} = \frac{a-b}{a}$

$i = (-1)^{1/2} \qquad \varphi + 1 = \varphi^2$

$\varphi_i = (-1)^{1/3} \qquad \varphi - 1 = \frac{1}{\varphi}$

$i = e^{\pi i/2} \qquad \varphi_i - 1 = -\frac{1}{\varphi_i}$

$\varphi_i = e^{\pi i/3}$

$\varphi = e^{\pi i/5} + e^{-\pi i/5}$

$\sqrt{5} = \frac{1}{\varphi} + \varphi$

$F(n) = \frac{\varphi^n - (-\varphi)^{-n}}{\sqrt{5}}$

$\varphi = \lim_{n\to\infty}\frac{F_{n+1}}{F_n}$

$\varphi^n = \varphi^{n-1} + \varphi^{n-2}$

$\varphi_i^n = \varphi_i^{n-1} - \varphi_i^{n-2}$

$\varphi_i = \frac{\omega_1}{\omega_2}$

$Re(\omega_1) = \frac{1}{2}\omega_2$

$Im(\omega_1) = \frac{\sqrt{3}}{2}\omega_2$

$\varphi = \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots}}}}$

$\varphi_i = \sqrt{-1 + \sqrt{-1 + \sqrt{-1 + \sqrt{-1 + \cdots}}}}$

$P = \sqrt[3]{1 + \sqrt[3]{1 + \sqrt[3]{1 + \sqrt[3]{1 + \cdots}}}}$

$\varphi = \sqrt{1^2 + \sqrt{1^2 + \sqrt{1^2 + \sqrt{1^2 + \cdots}}}}$

$\varphi_i = \sqrt{i^2 + \sqrt{i^2 + \sqrt{i^2 + \sqrt{i^2 + \cdots}}}}$

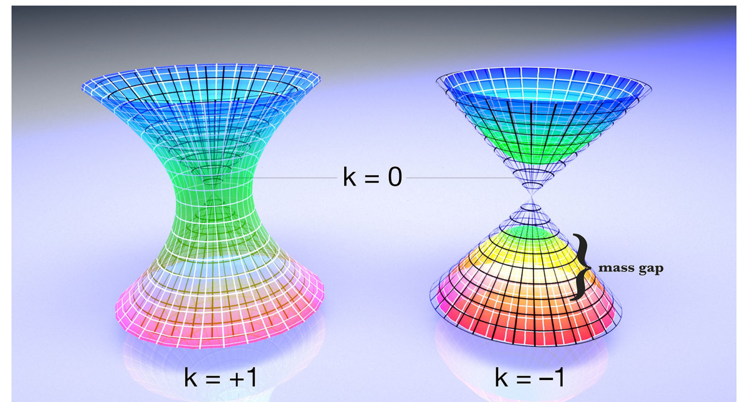
$\sin(x)$ = sine
$\cos(x)$ = cosine
$\tan(x)$ = tangent
$\cot(x)$ = cotangent
$\sinh(x)$ = hyperbolic sine
$\cosh(x)$ = hyperbolic cosine
$\coth(x)$ = hyperbolic cotangent
$\text{csch}(x)$ = hyperbolic cosecant
$\log(x)$ = natural logarithm
$Li_n(x)$ = polylogarithm of order n
$Q(a,x)$ = Q-regularized gamma function
$P(a,x)$ = P-regularized gamma function
$\Psi(x)$ = digamma function
$\Gamma(x)$ = gamma function
$n!$ = factorial
$!n$ = derangement
$\zeta(x)$ = Reimann zeta function
$B(x)$ = beta function
$Cl_s(x)$ = Clausen function
$K(i)$ = complete elliptic integral, 1st kind
$Im(x)$ = imaginary part of x
$Re(x)$ = real part of x
$\prod$ = product
$\sum$ = sum
$\int$ = integral
r_+ = proton radius
r_n = neutron radius

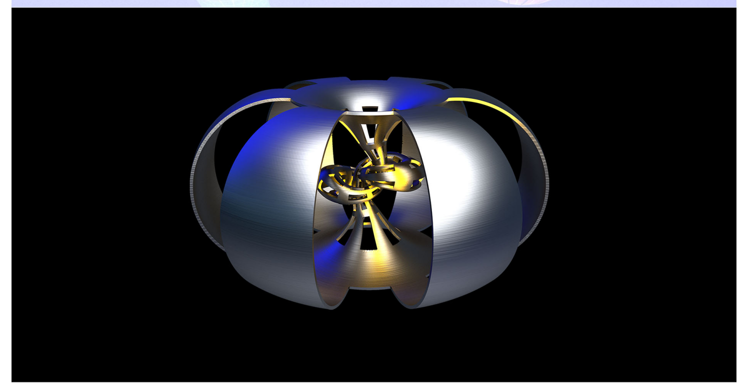
Physics Monastery
Thad Roberts
October 28, 2025

$$Cl_2(x) = \sum_{n=1}^{\infty}\frac{\sin(nx)}{n^2} = \frac{\sin(1x)}{1^2} + \frac{\sin(2x)}{2^2} + \frac{\sin(3x)}{3^2} + \frac{\sin(4x)}{4^2} + \frac{\sin(5x)}{5^2} + \cdots$$

$Cl_2\left(2\pi n \pm \frac{\pi}{3}\right) = \pm G_{\text{Gi}} \qquad Cl_2\left(2\pi n \pm \frac{2\pi}{3}\right) = \pm\frac{2G_{\text{Gi}}}{3} \qquad \frac{d}{dx}Cl_2(x) = Cl_1(x)$

$Cl_2(n\pi) = 0 \qquad \max = G_{\text{Gi}} \qquad \min = -G_{\text{Gi}} \qquad Cl_2(-x) = -Cl_2(x)$





legend

$t_p = 5.39125836832313\ldots\times 10^{-44}$ s	Planck time
$l_p = 1.61625918175645\ldots\times 10^{-35}$ m	Planck length
$q_p = 1.87554596713962\ldots\times 10^{-18}$ C	Planck charge
$T_p = 1.41678698590795\ldots\times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579\ldots\times 10^{-8}$ kg	Planck mass
5	number of dimensions
7	break in scale symmetry
$ж_1 = 0.0854245431533304\ldots$	1st hyperbolic partition constant
$ж_2 = 3.66756753485501\ldots$	2nd hyperbolic partition constant
$ж_3 = -1.87649603900417\ldots + 4.06615262615972\ldots i$	3rd hpc
$ж_4 = -1.87649603900417\ldots - 4.06615262615972\ldots i$	4th hpc
$ж_r = 4.47826244916751\ldots$	hyperbolic radius constant
$ж_\theta = 2.00316562310924\ldots$	hyperbolic radian constant
$G_{\text{Gi}} = 1.01494160640965\ldots$	Gieseking's constant
$V_{\text{fe}} = 2.02988321281930\ldots$	figure eight knot hyperbolic volume
$K = 0.9159655941772190\ldots$	Catalan's constant
$C_d = 0.291560904030818\ldots$	domino tiling constant
$D_d = 1.79162281206959\ldots$	dimer constant
$\omega_1 = 0.764977018528596\ldots + 1.32497062714087\ldots i$	omega_1 constant
$\omega_2 = 1.529954037057192\ldots$	omega_2 constant
$i^{i^{i^{\cdot^{\cdot}}}} = 0.438282936727032\ldots + 0.360592471871385\ldots i$	power tower of i
$i = (-1)^{1/2}$	imaginary unit
$\varphi_i = (-1)^{1/3}$	imaginary golden ratio
$\varphi = 1.61803398874989\ldots$	golden ratio
$\psi = 1.46557123187676\ldots$	supergolden ratio
$\pi = 3.14159265358979\ldots$	Archimedes' constant
$P_{up} = 2.29558714939263\ldots$	universal parabolic constant
$s = 5.24411510858423\ldots$	arc length of the unit lemniscate
$L = 2.622057554292119\ldots$	lemniscate constant
$L_1 = 1.31102877714605\ldots$	1st lemniscate constant
$L_2 = 0.599070117367796\ldots$	2nd lemniscate constant
$C_U = 0.847213084793979\ldots$	ubiquitous constant
$G_{Ga} = 0.834626841674073\ldots$	Gauss's constant
$C_{\text{CF}} = 0.697774657964819\ldots$	continued fraction constant
$C_{\text{R1}} = 0.998136044598509\ldots$	Ramanujan's 1st continued fraction constant
$M = -1.74756459463318\ldots$	Madelung constant
$e = 2.71828182845904\ldots$	Euler's number
$\gamma = 0.577215664901532\ldots$	Euler-Mascheroni constant
$\Omega = 0.567143290409783\ldots$	omega constant
$S = 0.822825249678847\ldots$	Sierpiński's constant
$j_{0,1} = 2.404825557695772\ldots$	1st root of the Bessel function
$C_{\text{CFP}} = 1.19967864025773\ldots$	Real fixed point of the hyperbolic cotangent
$D_{\text{Do}} = 0.739085133215160\ldots$	Dottie number
$L_{\text{LL}} = 0.662743419349181\ldots$	Laplace limit
$\alpha_F = 2.50290787509589\ldots$	alpha Feigenbaum constant
$\delta_F = 4.66920160910299\ldots$	delta Feigenbaum constant
$\mathcal{S} = 3.24697960371714\ldots$	silver constant
$P = 1.32471795724474\ldots$	plastic constant
$C_Q = 0.605443657196732\ldots$	QRS constant
$C_{QA} = 2.03816937970215\ldots$	associated QRS constant

with:

Angela Arvizu, Vanessa Moon, Matthew Fox, David Heggli, Eschelon Azha, Avi Rubin, Wayne Eskridge, Dan Whittaker, Kevin Sirios, Mike Ritter, Jerry Gardner, Shauna Montgomery, WolframAlpha, NIST/CODATA